

# Investigating Windows Networking from Configuration to Support Handoff

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I completed five Windows diagnostics labs in a university-provided Azure VM. I worked from host and adapter settings through DNS, DHCP, processes, network endpoints, routes and firewall state, then collected the output into a support record.

## Tools and environment

Windows VM in Azure; Command Prompt; ipconfig, ping, nslookup, tasklist, netstat, netsh, tracert, pathping, arp, nbtstat; Notepad.

Individual coursework by Adnan Khan. May-June 2025. The VM was provided for the labs.

## Establish the workstation and network baseline

I checked the host and Ethernet 2 adapter before interpreting any failed network test. The configuration records a Microsoft Hyper-V adapter, IPv4 10.0.0.176 with a /20 subnet, gateway 10.0.0.1, and DNS/DHCP server 168.63.129.16.

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.22000.2538]
(c) Microsoft Corporation. All rights reserved.

C:\Users\data>IPCONFIG /all

Windows IP Configuration

Host Name . . . . . : labDMO1CB
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . :

Ethernet adapter Ethernet 2:

Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Hyper-V Network Adapter #2
Physical Address. . . . . :
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : ::5a4c:69f8:4fcb:4397%5(Preferred)
IPv4 Address. . . . . : 10.0.0.176(Preferred)
Subnet Mask . . . . . : 255.255.240.0
Lease Obtained. . . . . : Monday, May 12, 2025 4:47:13 AM
Lease Expires . . . . . : Thursday, June 18, 2161 11:37:38 AM
Default Gateway . . . . . : 10.0.0.1
DHCP Server . . . . . : 168.63.129.16
DHCPv6 IAID . . . . . :
DHCPv6 Client DUID. . . . . :
DNS Servers . . . . . : 168.63.129.16
NetBIOS over Tcpip. . . . . : Enabled

C:\Users\data>
```

Network baseline. Source portfolio p. 5.

```
C:\Users\data>HOSTNAME
labDMO1CB

C:\Users\data>
```

Identifying the workstation. Source portfolio p. 8.

## Separate name resolution from reachability

I pinged a public hostname and received timeouts after the name resolved. I then used nslookup to query DNS directly. The returned address supports a working lookup; the missing echo replies leave the remote ICMP path unconfirmed.

```
C:\Users\data>www.ryerson.ca
'www.ryerson.ca' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\data>Ping www.ryerson.ca.

Pinging www.ryerson.ca [141.117.126.20] with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 141.117.126.20:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\data>
```

Testing a public hostname. Source portfolio p. 6.

```
C:\Users\data>nslookup www.ryerson.ca
Server:    UnKnown
Address:   168.63.129.16

Non-authoritative answer:
Name:     www.ryerson.ca
Address:  141.117.126.20

C:\Users\data>_
```

Separating dns from ping. Source portfolio p. 7.

## Read aliases and service-related DNS records

I extended the DNS checks to aliases, MX records and an SRV query. These queries ask different questions: the address behind a name, mail destinations, or service records. I read the returned record type rather than treating every answer as proof that an application was reachable.

```
C:\Users\data>nslookup www.microsoft.com
Server:   UnKnown
Address:  168.63.129.16

Non-authoritative answer:
Name:     e13678.dscb.akamaiedge.net
Addresses: 2600:140a:5000:986::356e
           2600:140a:5000:987::356e
           23.53.170.101
Aliases:  www.microsoft.com
           www.microsoft.com-c-3.edgekey.net
           www.microsoft.com-c-3.edgekey.net.globalredir.akadns.net

C:\Users\data>
```

Reading dns aliases. Source portfolio p. 18.

```
C:\Users\data>nslookup -type=MX gmail.com
Server:   UnKnown
Address:  168.63.129.16

Non-authoritative answer:
gmail.com      MX preference = 5, mail exchanger = gmail-smtp-in.l.google.com
gmail.com      MX preference = 40, mail exchanger = alt4.gmail-smtp-in.l.google.com
gmail.com      MX preference = 20, mail exchanger = alt2.gmail-smtp-in.l.google.com
gmail.com      MX preference = 30, mail exchanger = alt3.gmail-smtp-in.l.google.com
gmail.com      MX preference = 10, mail exchanger = alt1.gmail-smtp-in.l.google.com

gmail-smtp-in.l.google.com      internet address = 142.250.111.26
```

Querying mail records. Source portfolio p. 49.

# Inspect processes and their hosted services

I used tasklist to inventory processes and tasklist /svc to see the services associated with them. This connects a process identifier to service ownership and provides context before investigating a listening port or a busy process.

```
C:\Users\data>tasklist
```

| Image Name          | PID  | Session Name | Session# | Mem Usage |
|---------------------|------|--------------|----------|-----------|
| System Idle Process | 0    | Services     | 0        | 8 K       |
| System              | 4    | Services     | 0        | 144 K     |
| Registry            | 116  | Services     | 0        | 14,812 K  |
| smss.exe            | 472  | Services     | 0        | 1,276 K   |
| csrss.exe           | 600  | Services     | 0        | 6,100 K   |
| wininit.exe         | 672  | Services     | 0        | 7,132 K   |
| csrss.exe           | 692  | Console      | 1        | 5,388 K   |
| winlogon.exe        | 772  | Console      | 1        | 10,880 K  |
| services.exe        | 816  | Services     | 0        | 9,656 K   |
| lsass.exe           | 836  | Services     | 0        | 18,420 K  |
| svchost.exe         | 960  | Services     | 0        | 29,716 K  |
| fontdrvhost.exe     | 988  | Services     | 0        | 3,800 K   |
| fontdrvhost.exe     | 992  | Console      | 1        | 3,472 K   |
| svchost.exe         | 560  | Services     | 0        | 14,472 K  |
| svchost.exe         | 848  | Services     | 0        | 11,604 K  |
| LogonUI.exe         | 1068 | Console      | 1        | 50,648 K  |
| dwm.exe             | 1076 | Console      | 1        | 45,220 K  |
| svchost.exe         | 1100 | Services     | 0        | 7,636 K   |
| svchost.exe         | 1112 | Services     | 0        | 110,188 K |
| svchost.exe         | 1264 | Services     | 0        | 5,552 K   |
| svchost.exe         | 1272 | Services     | 0        | 9,980 K   |
| svchost.exe         | 1296 | Services     | 0        | 12,408 K  |

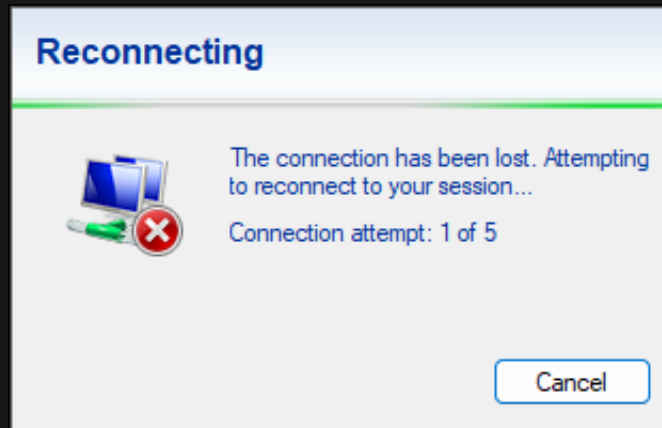
Building a process inventory. Source portfolio p. 19.

|                      |      |                      |
|----------------------|------|----------------------|
| WmiPrvSE.exe         | 4600 | N/A                  |
| WmiPrvSE.exe         | 4884 | N/A                  |
| svchost.exe          | 5020 | wlidsvc              |
| svchost.exe          | 4844 | RmSvc                |
| svchost.exe          | 4404 | UsoSvc               |
| svchost.exe          | 5856 | wuauserv             |
| taskhostw.exe        | 5964 | N/A                  |
| svchost.exe          | 4956 | pla                  |
| NisSrv.exe           | 5544 | WdNisSvc             |
| svchost.exe          | 5728 | BITS                 |
| svchost.exe          | 5780 | SSDPSSRV             |
| MaSecAgentProv.exe   | 5272 | N/A                  |
| conhost.exe          | 5936 | N/A                  |
| LabServicesAgent.exe | 5292 | LabServicesIdleAgent |
| WmiPrvSE.exe         | 5380 | N/A                  |
| svchost.exe          | 1328 | AppXSvc              |
| svchost.exe          | 6388 | DoSvc                |
| svchost.exe          | 6436 | StorSvc              |
| svchost.exe          | 6492 | ClipSVC              |
| csrss.exe            | 7028 | N/A                  |
| winlogon.exe         | 7108 | N/A                  |
| fontdrvhost.exe      | 7156 | N/A                  |
| dwm.exe              | 1132 | N/A                  |
| WUDFHost.exe         | 1564 | N/A                  |
| svchost.exe          | 2208 | ScDeviceEnum         |
| rdpclip.exe          | 5336 | N/A                  |
| sihost.exe           | 3176 | N/A                  |

Mapping processes to services. Source portfolio p. 24.

## Check DHCP renewal after losing remote access

The DHCP release exercise interrupted the remote session. I recorded that symptom, then checked `ipconfig /renew`. The renewed adapter output again shows 10.0.0.176 and gateway 10.0.0.1. This is a recovery configuration check, not a diagnosis of every connectivity issue.



Dhcp release and remote access. Source portfolio p. 30.

```
C:\Users\data>ipconfig /renew
```

```
Windows IP Configuration
```

```
Ethernet adapter Ethernet 2:
```

```
Connection-specific DNS Suffix . : 
Link-local IPv6 Address . . . . . : ::1
IPv4 Address. . . . . : 10.0.0.176
Subnet Mask . . . . . : 255.255.240.0
Default Gateway . . . . . : 10.0.0.1
```

```
C:\Users\data>
```

Inspecting the renewed lease. Source portfolio p. 31.

## Inspect, clear and revisit the DNS cache

I inspected cached records with `ipconfig /displaydns`, cleared the resolver cache with `ipconfig /flushdns`, and reviewed later records after browsing. The flush command returned a success message. Later cache entries show that DNS information was collected again.

- I also practised tree for directory structure and console colour commands; those supporting exercises do not need repeated screenshots.

```
C:\Users\data>ipconfig /flushdns

Windows IP Configuration

Successfully flushed the DNS Resolver Cache.

C:\Users\data>_
```

Clearing the resolver cache. Source portfolio p. 42.

```
Record Name . . . . . : prdv4a.aadg.msidentity.com
Record Type . . . . . : 5
Time To Live . . . . . : 131
Data Length . . . . . : 8
Section . . . . . : Answer
CNAME Record . . . . . : www.tm.v4.a.prd.aadg.trafficmanager.net

Record Name . . . . . : www.tm.v4.a.prd.aadg.trafficmanager.net
Record Type . . . . . : 1
Time To Live . . . . . : 131
Data Length . . . . . : 4
Section . . . . . : Answer
A (Host) Record . . . . : 40.126.29.6

Record Name . . . . . : www.tm.v4.a.prd.aadg.trafficmanager.net
Record Type . . . . . : 1
Time To Live . . . . . : 131
Data Length . . . . . : 4
Section . . . . . : Answer
A (Host) Record . . . . : 40.126.29.11
```

Observing cache repopulation. Source portfolio p. 43.

## Connect network endpoints to processes

I used `netstat -a` to inspect listeners and connections, then `netstat -o` to include process IDs. I also inspected TCP connections, UDP endpoints and the dynamic TCP port range with `netsh`. These outputs describe endpoints and allocation settings at the time of the lab.

```
C:\Users\data>netstat -o

Active Connections

Proto Local Address          Foreign Address         State       PID
TCP   10.0.0.176:3389         pool-99-234-219-184:58210 ESTABLISHED 1136
TCP   10.0.0.176:49691       168.63.129.16:32526    ESTABLISHED 3548
TCP   10.0.0.176:49701       168.63.129.16:http     ESTABLISHED 3548
TCP   10.0.0.176:49753       168.63.129.16:32526    ESTABLISHED 3400
TCP   10.0.0.176:49793       20.189.173.18:https     ESTABLISHED 6604
TCP   10.0.0.176:49913       172.172.255.216:https   ESTABLISHED 3600
TCP   10.0.0.176:50055       172.172.255.216:https   ESTABLISHED 3600
TCP   10.0.0.176:50119       40.117.80.207:https     ESTABLISHED 6604
TCP   10.0.0.176:50127       20.150.78.196:https     TIME_WAIT   0
TCP   10.0.0.176:50129       52.182.141.63:https     TIME_WAIT   0
TCP   10.0.0.176:50131       150.171.28.11:https     ESTABLISHED 9608
TCP   10.0.0.176:50160       a23-222-17-172:https    CLOSE_WAIT  7276
TCP   10.0.0.176:50161       a23-222-17-172:https    CLOSE_WAIT  7276
TCP   10.0.0.176:50162       13.107.226.254:https    CLOSE_WAIT  7276
TCP   10.0.0.176:50163       a23-5-103-252:http      ESTABLISHED 7276
TCP   10.0.0.176:50164       a23-222-17-172:https    CLOSE_WAIT  7276
TCP   10.0.0.176:50165       a23-222-17-172:https    CLOSE_WAIT  7276
TCP   10.0.0.176:50166       a23-222-17-172:https    CLOSE_WAIT  7276
TCP   10.0.0.176:50198       4.155.164.36:https      TIME_WAIT   0
TCP   10.0.0.176:50201       a2-16-170-214:http      TIME_WAIT   0
TCP   10.0.0.176:50202       151.101.126.172:http    TIME_WAIT   0
TCP   10.0.0.176:50203       151.101.126.172:http    TIME_WAIT   0
TCP   10.0.0.176:50229       20.189.173.28:https     ESTABLISHED 6604
TCP   10.0.0.176:50236       52.123.251.148:https    TIME_WAIT   0
```

Associating connections with pids. Source portfolio p. 55.

```
netsh interface ipv4>show dynamicportrange protocol=tcp

Protocol tcp Dynamic Port Range
-----
Start Port      : 49152
Number of Ports : 16384

netsh interface ipv4>
```

Checking dynamic tcp ports. Source portfolio p. 68.

## Check the local stack and compare IP versions

IPv4 loopback and IPv6 localhost each returned four replies with zero packet loss. I kept those local checks separate from traceroute tests: the IPv4 traces had limited replies, while the IPv6 trace could not resolve its target.

```
C:\Users\data>ping 127.0.0.1

Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 127.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Validating ipv4 loopback. Source portfolio p. 59.

```
C:\Users\data>ping localhost

Pinging labDM01CB [::1] with 32 bytes of data:
Reply from ::1: time<1ms
Reply from ::1: time<1ms
Reply from ::1: time<1ms
Reply from ::1: time<1ms

Ping statistics for ::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Validating localhost over ipv6. Source portfolio p. 60.

## Review neighbors, routes and device identity

I inspected ARP and IPv4 neighbor information, then reviewed the route table and assigned DNS servers. I also used getmac, hostname and nbtstat for device identity. The default route identifies the gateway; a neighbor entry is a local mapping rather than a full remote path.

```
netsh interface ipv4>show neighbors

Interface 1: Loopback Pseudo-Interface 1

Internet Address          Physical Address  Type
-----
224.0.0.22                Permanent
239.255.255.250           Permanent

Interface 5: Ethernet 2

Internet Address          Physical Address  Type
-----
10.0.0.1                  Reachable
10.0.15.255              ff-ff-ff-ff-ff-ff Permanent
224.0.0.22               01-00-5e-00-00-16 Permanent
224.0.0.251              01-00-5e-00-00-fb Permanent
224.0.0.252              01-00-5e-00-00-fc Permanent
239.255.255.250          01-00-5e-7f-ff-fa Permanent
255.255.255.255          ff-ff-ff-ff-ff-ff Permanent

netsh interface ipv4>
```

Inspecting ipv4 neighbors. Source portfolio p. 62.

```
netsh interface ipv4>show route

Publish  Type      Met  Prefix          Idx  Gateway/Interface Name
-----
No       Manual    0    0.0.0.0/0       5    10.0.0.1
No       System    256  10.0.0.0/20     5    Ethernet 2
No       System    256  10.0.0.176/32   5    Ethernet 2
No       System    256  10.0.15.255/32  5    Ethernet 2
No       System    256  127.0.0.0/8     1    Loopback Pseudo-Interface 1
No       System    256  127.0.0.1/32    1    Loopback Pseudo-Interface 1
No       System    256  127.255.255.255/32 1    Loopback Pseudo-Interface 1
No       Manual    1    168.63.129.16/32 5    10.0.0.1
No       Manual    1    169.254.169.254/32 5    10.0.0.1
No       System    256  224.0.0.0/4     1    Loopback Pseudo-Interface 1
No       System    256  224.0.0.0/4     5    Ethernet 2
No       System    256  255.255.255.255/32 1    Loopback Pseudo-Interface 1
No       System    256  255.255.255.255/32 5    Ethernet 2

netsh interface ipv4>
```

Inspecting forwarding decisions. Source portfolio p. 63.

# Inspect firewall state and preserve the investigation

I recorded the firewall state and port-exception views using the commands provided in the labs. Those legacy netsh firewall commands display deprecation warnings. I then appended hostname, route, interface and connection output to Q1.txt and opened it in Notepad for a support handoff.

- The collection uses >> to append. The pathping command appears separately in the source, so I do not claim its output was saved to Q1.txt.

```
netsh firewall>show state

Firewall status:
-----
Profile                               = Standard
Operational mode                       = Enable
Exception mode                         = Enable
Multicast/broadcast response mode     = Enable
Notification mode                     = Enable
Group policy version                  = Windows Defender Firewall
Remote admin mode                     = Disable

Ports currently open on all network interfaces:
Port  Protocol Version Program
-----
No ports are currently open on all network interfaces.

IMPORTANT: Command executed successfully.
However, "netsh firewall" is deprecated;
use "netsh advfirewall firewall" instead.
For more information on using "netsh advfirewall firewall" commands
instead of "netsh firewall", see KB article 947709
at https://go.microsoft.com/fwlink/?linkid=121488 .

netsh firewall>
```

Reviewing firewall state. Source portfolio p. 72.

```
lab0001C8
=====
Interface List
5...60 45 bd 5f f9 14 ....Microsoft Hyper-V Network Adapter #2
1...00 00 00 00 00 00 ....Software Loopback Interface 1
=====

IPv4 Route Table
=====
Active Routes:
Network Destination  Netmask          Gateway          Interface        Metric
0.0.0.0              0.0.0.0          10.0.0.1         10.0.0.176       10
10.0.0.0             255.255.240.0    On-link          10.0.0.176       266
10.0.0.176           255.255.255.255  On-link          10.0.0.176       266
127.0.0.0            255.0.0.0        On-link          127.0.0.1        331
127.0.0.1            255.255.255.255  On-link          127.0.0.1        331
127.255.255.255      255.255.255.255  On-link          127.0.0.1        331
168.63.129.16        255.255.255.255  10.0.0.1         10.0.0.176       11
169.254.169.254      255.255.255.255  10.0.0.1         10.0.0.176       11
224.0.0.0            240.0.0.0        On-link          127.0.0.1        331
224.0.0.0            240.0.0.0        On-link          10.0.0.176       266
255.255.255.255      255.255.255.255  On-link          127.0.0.1        331
255.255.255.255      255.255.255.255  On-link          10.0.0.176       266
=====
Persistent Routes:
None

IPv6 Route Table
=====
Active Routes:
If Metric Network Destination  Gateway
1 331 ::1/128 On-link
5 266 fe80::/64 On-link
5 266 ::ac4:69f8:4fcb:4397/128 On-link
1 331 ff00::/8 On-link
5 266 ff00::/8 On-link
=====
Persistent Routes:
None
lab0001C8
Interface Statistics
```

Collecting a diagnostic report. Source portfolio p. 73.

## Put the troubleshooting findings together

The evidence shows valid addressing, successful DNS answers and working loopback tests alongside incomplete ping and trace results. I narrowed the problem without declaring a root cause that the tests did not establish. A next investigation would test the affected application port and review the applicable network policy.

```
Microsoft Windows [Version 10.0.22000.2538]
(c) Microsoft Corporation. All rights reserved.

C:\Users\data>netstat -r >> Q1.txt

C:\Users\data>pathping jupiter.scs.ryerson.ca

Tracing route to jupiter.scs.ryerson.ca [141.117.57.31]
over a maximum of 30 hops:
  0  labDMO1CB.tq4dk2uwsfxulde2izwg0pn1pf [10.0.0.176]
  1  * * * *
Computing statistics for 0 seconds...
   Source to Here   This Node/Link
Hop  RTT      Lost/Sent = Pct  Lost/Sent = Pct  Address
  0          0/0 = 0%          0/0 = 0%      labDMO1CB [10.0.0.176]

Trace complete.

C:\Users\data>hostname >> Q1.txt

C:\Users\data>netstat -e >> Q1.txt

C:\Users\data>netstat -p TCP >> Q1.txt

C:\Users\data>netstat -f >> Q1.txt

C:\Users\data>notepad Q1.txt

C:\Users\data>_
```

Recording the collection workflow. Source portfolio p. 75.

## What I learned

- Use separate tests for DNS, the local stack and the remote path.
- Connect endpoint information to process and service ownership.
- Record commands and observed results so another technician can continue the investigation.

Source: the full CITM301 portfolio supplied for this project. Evidence is from the original coursework; this edition adds no new lab test results. Repeated captures and earlier drafts are consolidated. Revised design summaries are labelled, and sensitive identifiers are masked.